

Alex Economopoulos

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Skills

Languages: Go, C++, Python, JavaScript, Java, MATLAB

Frameworks/Libraries : Flask, Gunicorn, OpenGL, GLFW, ImGui, Tkinter, OpenCV, zxing-cpp

Tools: Git, CI/CD, Linux, Docker, Elastic Stack, Prometheus, Grafana, Journald, Certbot, Bash

Systems & Protocols: HTTP, TLS, SSH, TCP/IP, Multithreading

Experience

Student, Software Development

Bell Media, Scarborough, ON

May 2026 – August 2026

- Automated the end-to-end TLS certificate lifecycle, eliminating manual renewal, distribution, and monitoring across Linux infrastructure, saving **50+ hours** of manual effort annually
- Implemented zero-downtime TLS certificate hot reloading in a custom Go server by dynamically updating in-memory certificates leveraging Go's garbage collection to free up stale certificates, **eliminating node restarts and traffic removal during certificate rotation**
- Refactored a tightly coupled Go monitoring service into independent data streams, decoupling collectors and writers to isolate failures and enable stream-specific scrape intervals; implemented ring buffers, journal cursors, atomic persistence, and fault-tolerant recovery to prevent data loss, **reducing system-metric log volume by 60×**
- Expanded infrastructure observability and hardened monitoring reliability by adding hardware-failure, disk, TLS, and certificate-renewal status telemetry, resolving auditd-driven dmesg log flooding, and implementing threshold-based log rotation, reducing critical failure-identification latency from up to **12 hours to near-instant detection**

Guidance, Navigation, Controls Member

UW ORBITAL, Waterloo, ON

September 2025 – May 2026

- Simulated satellite eclipse periods using vector math and **Simulink** to predict solar panel charging capabilities
- Modelled expected Earth magnetic field measurements at the satellite using **MATLAB/Simulink** to support magnetorquer-based attitude stabilization
- Estimated Viscous and Coulomb friction coefficients to support **high fidelity reaction wheel model**

Engineering Intern

Hammerschlag and Joffe Engineering Consultancy, Toronto, ON

July – August 2024

- Updated fire protection layouts by mapping legacy devices for a large department store chain using **AutoCAD**
- Performed **electrical load and water usage calculations** for corporate clients to aid engineering analyses
- Reviewed tender documents and contractor submissions, ensuring accuracy and compliance with **Excel**
- Drafted and edited **shop drawing review reports** to support construction quality control

Projects

Graphing Calculator (C++)

- Developed a real-time mathematical function plotting application with **custom function parsing**
- Implemented zooming and panning using **coordinate transformations**
- Designed interactive **GUI** using **OpenGL** and **ImGui**

Groche (Python)

- Developed a full-stack grocery-to-recipe matching application using Python, Flask, JavaScript, HTML, and CSS, with **OpenCV** and **zxingcpp** to detect and decode grocery barcodes and build a digital representation of a user's cart
- Implemented **repeated greedy selection and fuzzy string matching** to map specific product descriptions to generalized recipe ingredients, selecting the closest matching recipe based on available ingredients
- Built an interactive recipe comparison interface that highlights missing and extra cart items, enabling users to identify ingredient gaps

2D Collision Simulator (Python)

- Simulated elastic and inelastic collisions with user-controlled mass, velocity, and position
- Applied conservation laws to compute post-collision states and energy loss
- Developed an interactive **GUI** using **Tkinter**

Education

University of Waterloo, ON

September 2025 – June 2030

Candidate for Bachelor of Applied Science in Biomedical Engineering

GPA: 3.94